

9T1: Spectral-based audio features

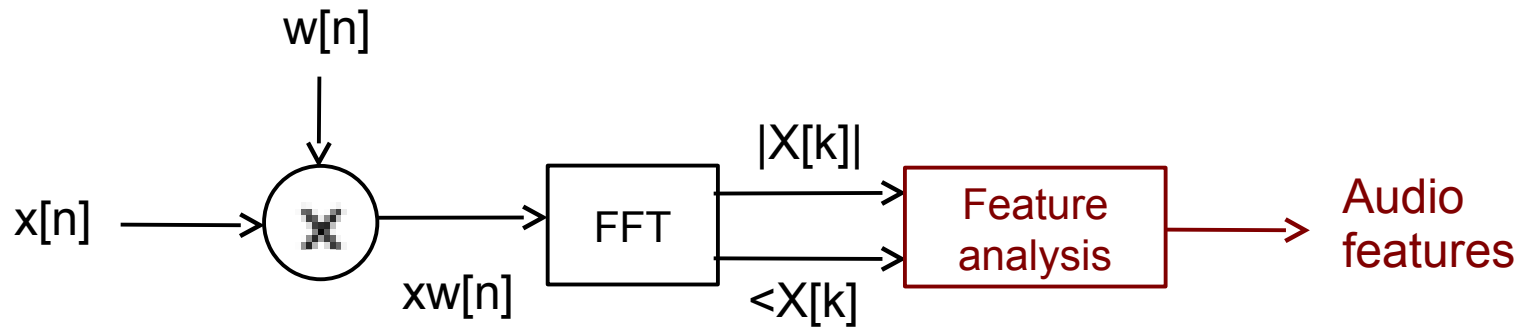
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- Introduction: audio features
- Single-frame spectral features
- Multiple-frames spectral features

Audio features



Essentia descriptors

- **Spectral descriptors:** BarkBands, MelBands, ERBBands, MFCC, GFCC, LPC, HFC, SpectralContrast, Inharmonicity and Dissonance, ...
- **Time-domain descriptors:** EffectiveDuration, ZCR, Loudness, ...
- **Tonal descriptors:** PitchSalienceFunction, PitchYinFFT, HPCP, TuningFrequency, Key, ChordsDetection, ...
- **Rhythm descriptors:** BeatTrackerDegara, BeatTrackerMultiFeature, BpMHistogramDescriptors, NoveltyCurve, OnsetDetection, Onsets, ...
- **SFX descriptors:** LogAttackTime, MaxToTotal, MinToTotal, TCToTotal, ...
- **High-level descriptors:** Danceability, DynamicComplexity, FadeDetection, SBic, ...

Single-frame spectral features

- Energy, RMS, Loudness
- Spectral centroid
- Mel-frequency cepstral coefficients (MFCC)
- Pitch salience
- Chroma (Harmonic pitch class profile, HPCP)

Energy, RMS, Loudness

Energy:

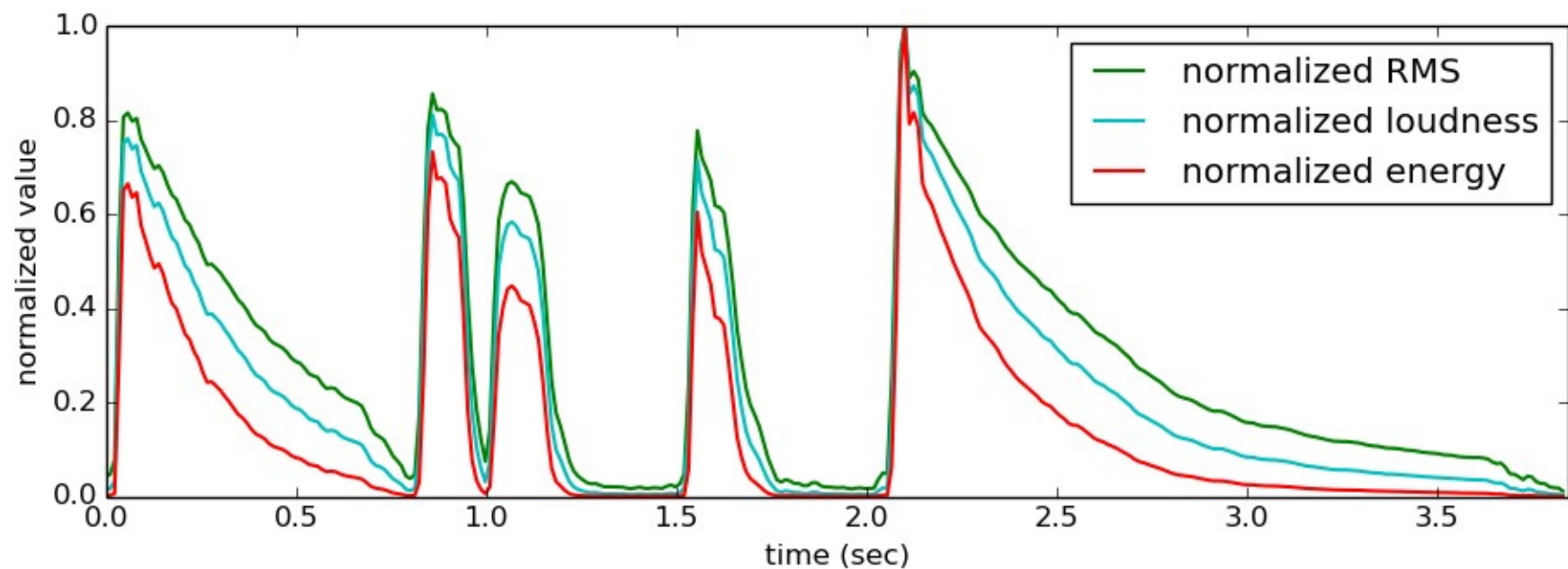
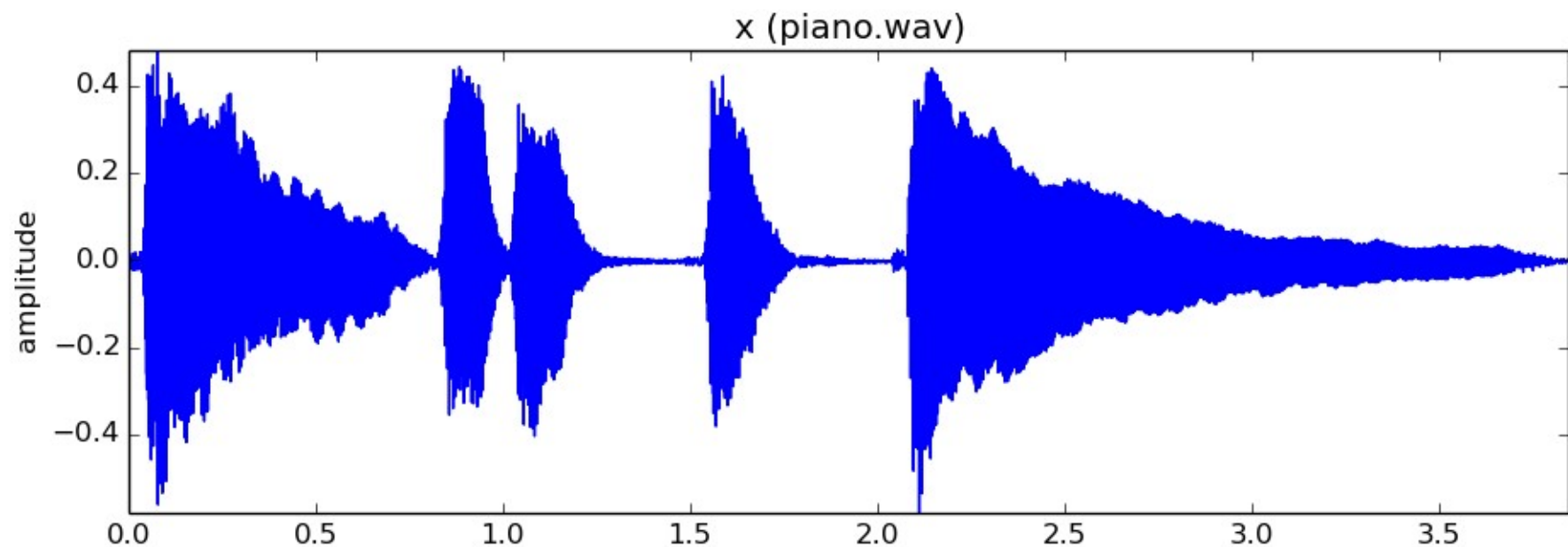
$$energy_l = \sum_{k=0}^{N-1} |X_l[k]|^2$$

Root mean square:

$$RMS_l = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} |X_l[k]|^2}$$

Steven's power law:

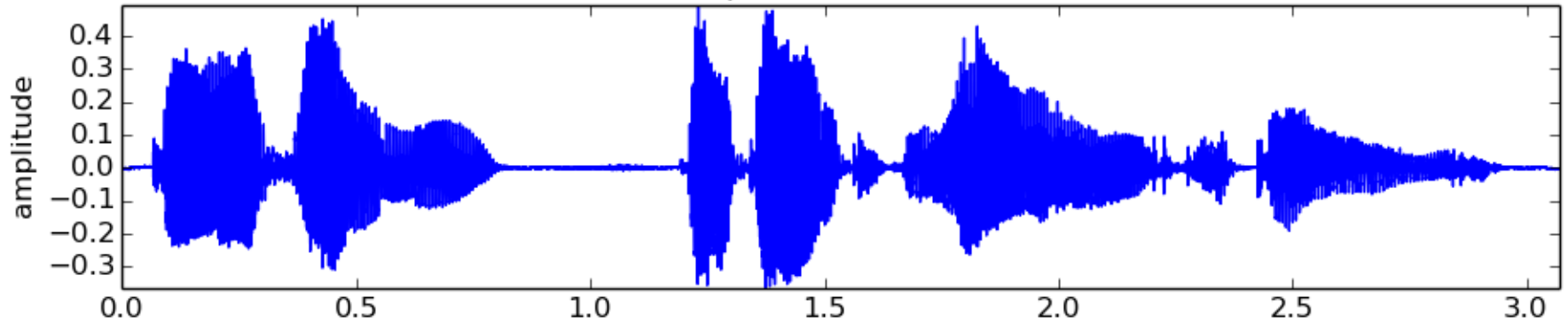
$$loudness_l = \left(\sum_{k=0}^{N-1} |X_l[k]|^2 \right)^{0.67}$$



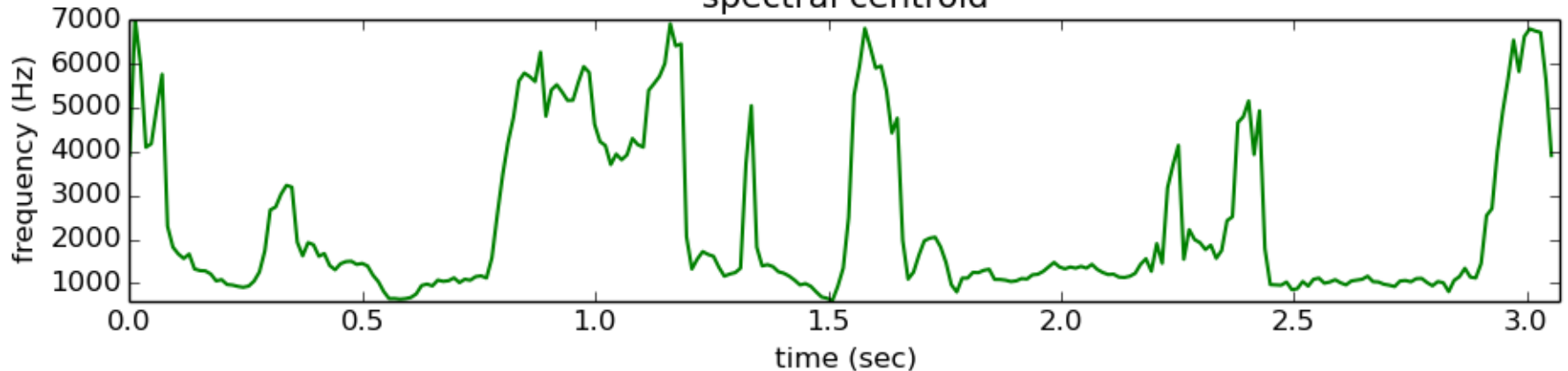
Spectral centroid

$$centroid_l = \frac{\sum_{k=0}^{N/2} k |X_l[k]|}{\sum_{k=0}^{N/2} |X_l[k]|}$$

x (speech-male.wav)



spectral centroid



Mel frequency cepstral coefficients

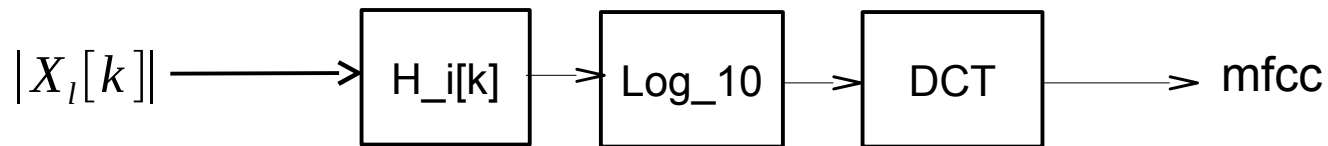
$$mfcc_l = DCT \left(\log_{10} \left(\sum_{k=0}^{N/2} |X_l[k]| H_i[k] \right) \right)$$

where

$|X[k]|$ is the positive magnitude spectrum

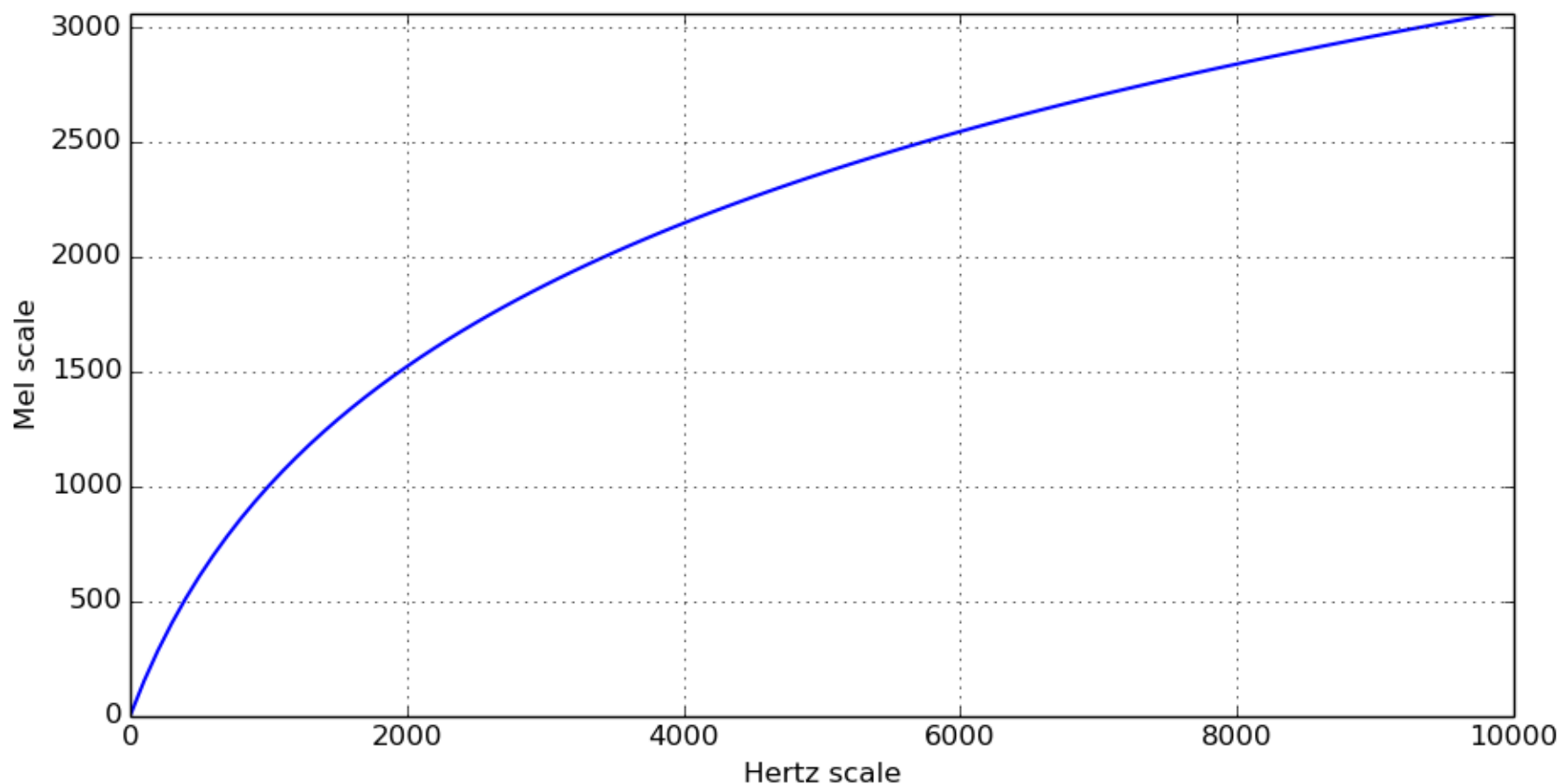
$H_i[k]$ is the mel scale filter bank for each filter i

$$DCT[m] (\text{Discrete Cosine Transform}) = \sum_{n=0}^{N-1} f[n] \cos \left(\frac{\pi}{N} \left(n + \frac{1}{2} \right) m \right)$$

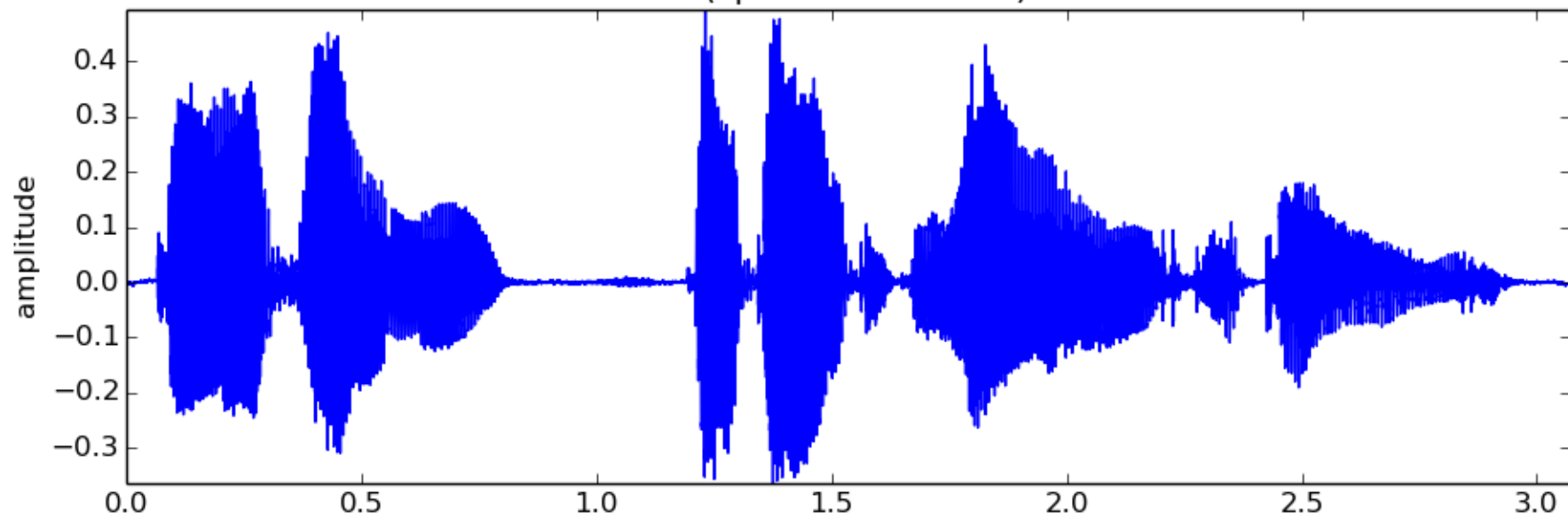


MFCC: Mel scale

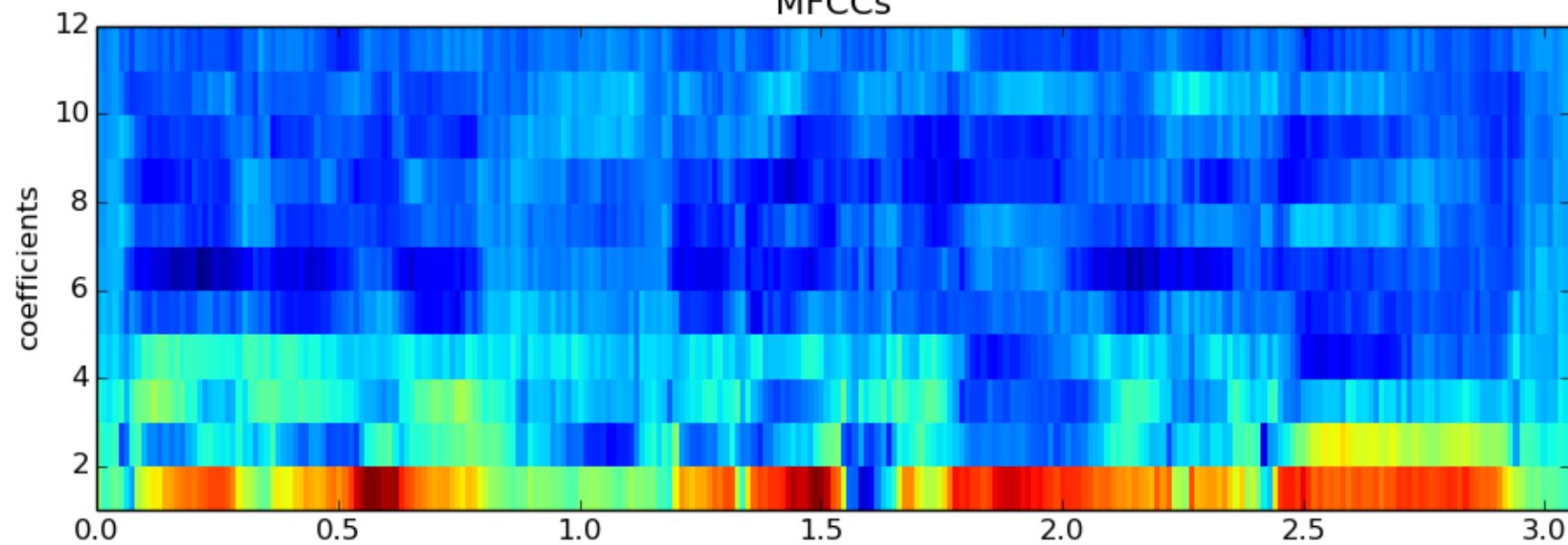
$$mel = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right)$$



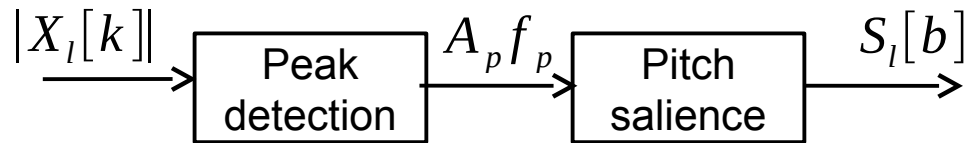
x (speech-male.wav)



MFCCs



Pitch salience



$$S[b] = \sum_{h=1}^H \sum_{p=1}^P e(A_p) g(b, h, f_p) (A_p)^\beta$$

where

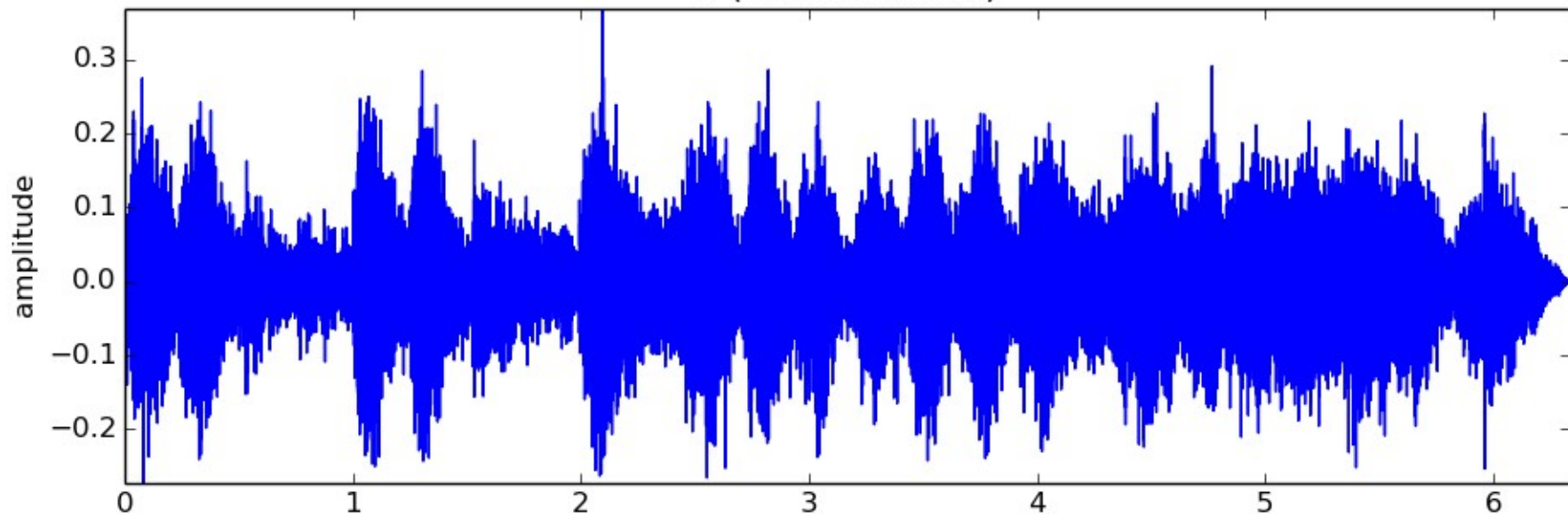
$S[b]$ = salience at bin frequency b (b expressed in cent scale)

$e()$ = magnitude threshold function

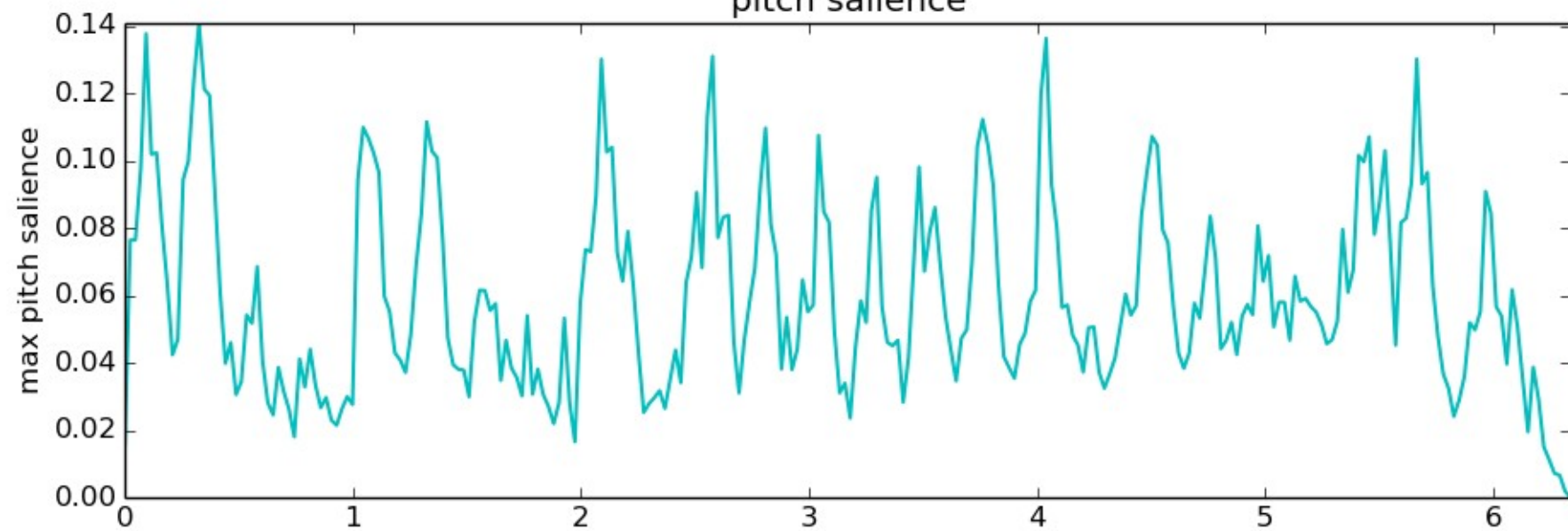
$g()$ = weighting function applied to peak p

β = magnitude compression value

x (orchestra.wav)



pitch salience



Chroma (Harmonic Pitch Class Profile)

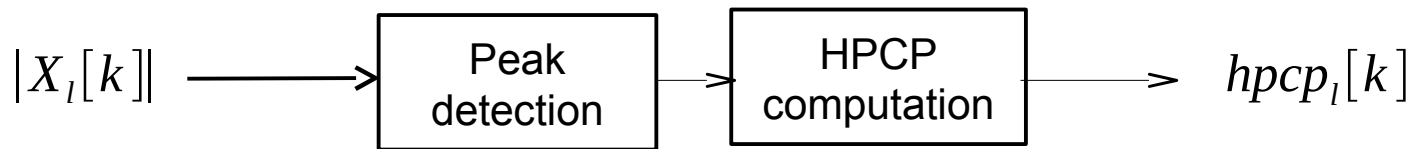
$$hpcp[k] = \sum_{p=1}^P w(k, f_p) A_p^2$$

where

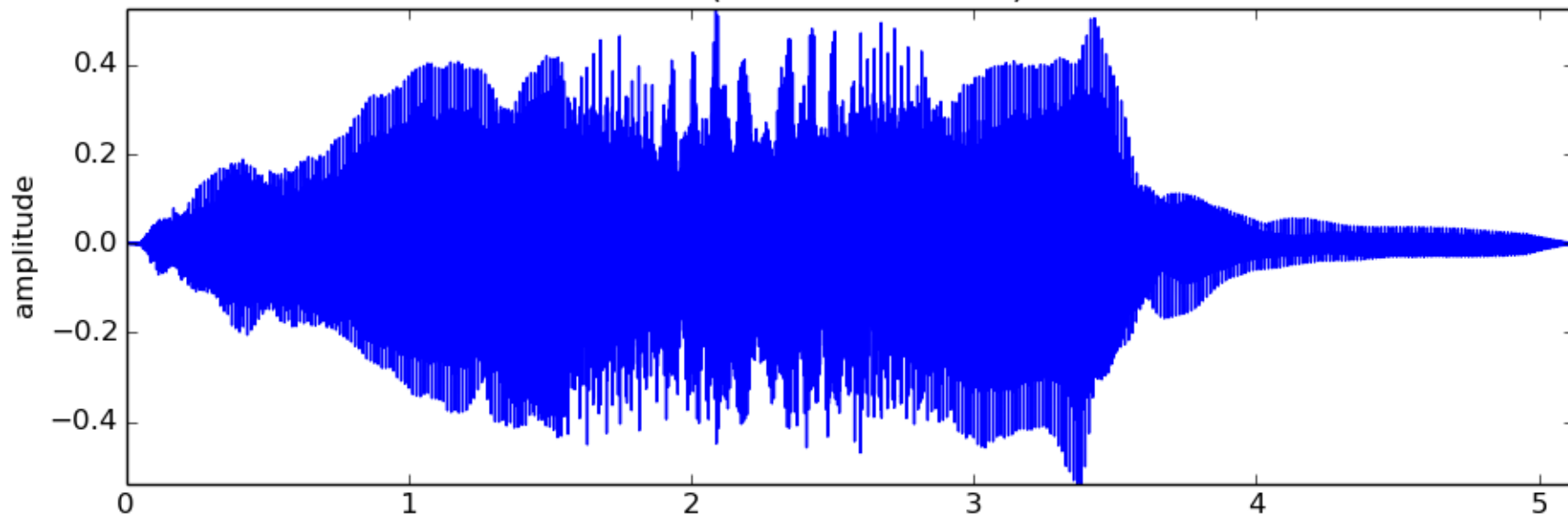
A_p = amplitude of spectral peak p

$w(k, f_p)$ = weight of the peak frequency f_p for bin k

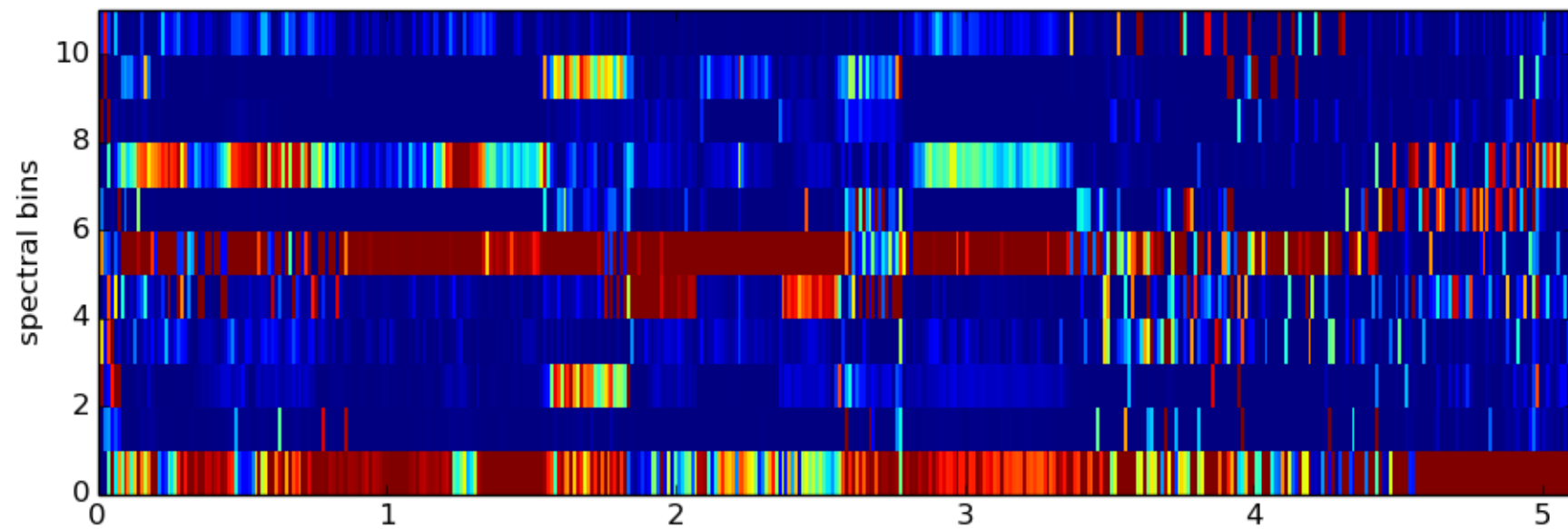
k = spectral bin locations of the chosen HPCP frequencies



x (cello-double.wav)



HPCP



Multiple-frames spectral features

- Event segmentation, onsets
- Predominant pitch
- Statistics of single-frame features

Event segmentation, onsets

- Spectral flux (used in segmentation)

$$SF_l = \sum_{k=0}^{N/2} H(|X_l[k]| - |X_{(l-1)}[k]|)$$

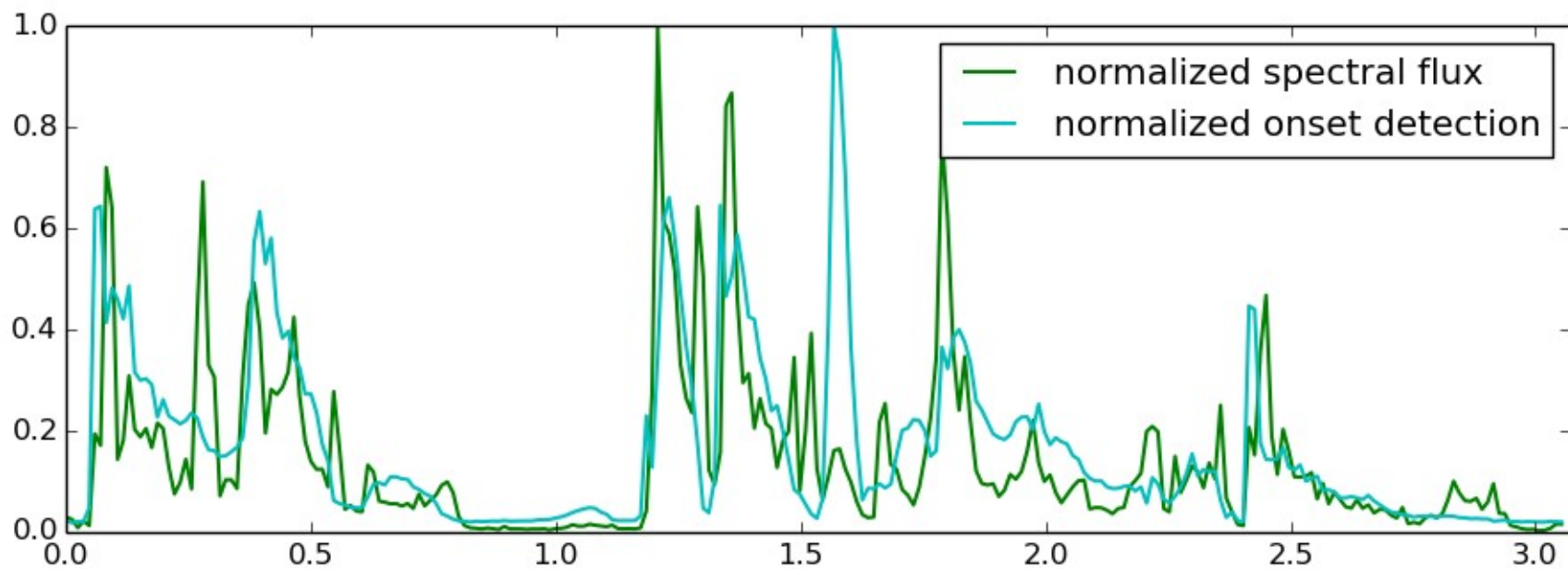
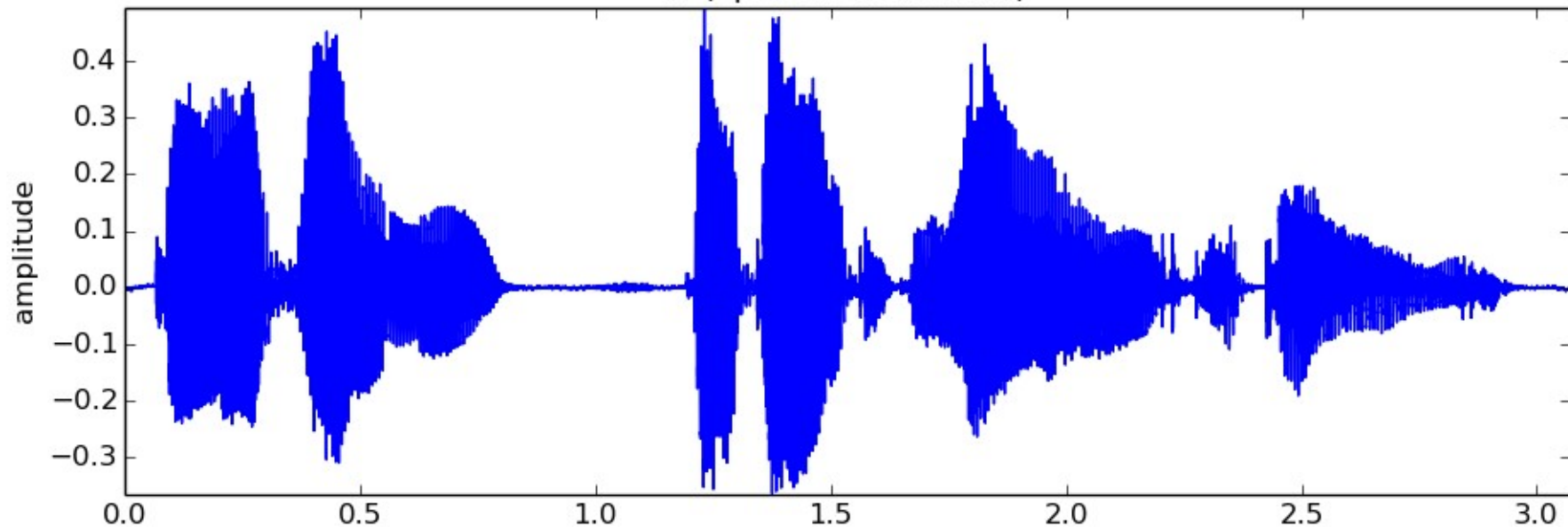
$$\text{where } H(x) = \frac{x + |x|}{2}$$

- Onset detection based on high-frequency content

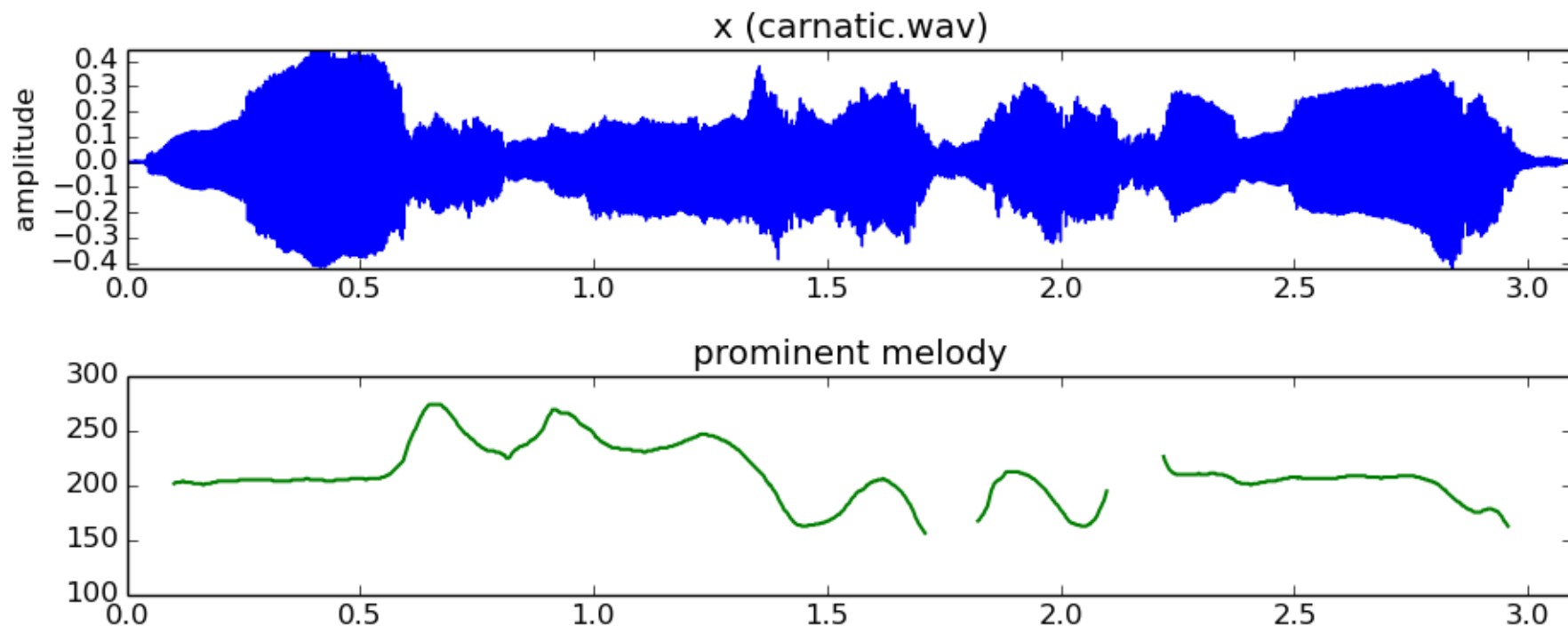
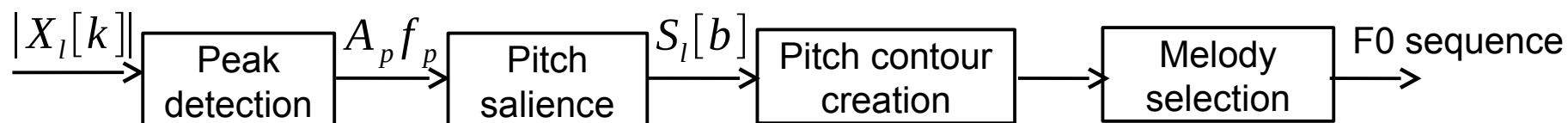
$$\text{Onset detection function} = HFC_l - HFC_{(l-1)}$$

$$\text{where } HFC_l = \sum_{k=1}^{N/2} |X_l[k]| k^2$$

x (speech-male.wav)



Predominant pitch



Statistics of single frame features

- Arithmetic mean (first moment)

$$mean = \frac{1}{N} \sum_{i=0}^{N-1} y[i]$$

- Variance (second moment)

$$variance = \frac{1}{N} \sum_{i=0}^{N-1} (y[i] - mean)^2$$

- Skewness (third moment)

$$skewness = \frac{\frac{1}{N} \sum_{i=0}^{N-1} (y[i] - mean)^3}{\left[\frac{1}{N-1} \sum_{i=0}^{N-1} (y[i] - mean)^2 \right]^{3/2}}$$

References

- Essentia: <http://essentia.upf.edu>
- http://en.wikipedia.org/wiki/Spectral_centroid
- http://en.wikipedia.org/wiki/Mel-frequency_cepstrum
- <http://en.wikipedia.org/wiki/Loudness>
- http://en.wikipedia.org/wiki/Harmonic_pitch_class_profiles
- [http://en.wikipedia.org/wiki/Onset_\(audio\)](http://en.wikipedia.org/wiki/Onset_(audio))
- [http://en.wikipedia.org/wiki/Moment_\(mathematics\)](http://en.wikipedia.org/wiki/Moment_(mathematics))
- Slides released under CC Attribution-Noncommercial-Share Alike license and code under Affero GPL license; available from <https://github.com/MTG/sms-tools>

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